

GVPT622 Lab Session

Probability Theory

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- ▶ Load the packages we'll use today and set the seed:

```
library(readxl)
library(tidyverse)
library(janitor)

set.seed(1234)
```

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Why `set.seed()`?

- ▶ In this session, we're working with randomness and chance
- ▶ `set.seed()` fixes the state of R's random number generator. The draws are still random, but they're the same random draws every time you run the script
- ▶ That's what makes random work replicable, so use it in your own research too

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 - ▷ Randomness
 - ▷ Independence
 - ▷ Law of Large Numbers
 - ▷ Conditional Probability
 - ▷ Frequentist and Bayesian perspectives

Randomness



Case Study: Whipping an Undecided Member

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- ▶ If you could predict the vote with certainty, you'd just write it down. You can't, so you want the outcome with the highest chance of happening
- ▶ Two steps:
 - ▷ List all the possible outcomes
 - ▷ Work out how likely each one is

- ▶ The first step is easy. The member votes yea or nay:

Possible Outcomes

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```
possible_outcomes <- c("YEA", "NAY")  
possible_outcomes
```

```
[1] "YEA" "NAY"
```

Possible Outcomes

- ▶ The first step is easy. The member votes yea or nay:

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possible_outcomes <- c("YEA", "NAY")  
possible_outcomes
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```
[1] "YEA" "NAY"
```

- ▶ How likely is each? One way to find out is through repeated trials. Watch this member cast many votes like this one and record what they do

- ▶ We'll simulate 10 votes, where yea and nay are equally likely:

Repeated Trials

- ▶ We'll simulate 10 votes, where yea and nay are equally likely:

```
repeat_trials <- sample(possible_outcomes, size = 10,  
                        replace = T, prob = c(0.5, 0.5))  
repeat_trials
```

```
[1] "NAY" "YEA" "YEA" "YEA" "YEA" "YEA" "NAY" "NAY" "YEA" "YEA"
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Repeated Trials

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- ▶ `prob = c(0.5, 0.5)` sets the true probability of each outcome.

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- ▶ Picture the names of the 13 members of the House Rules Committee in a hat. You draw a name and write it down
 - ▷ With replacement: put the name back before the next draw. You could draw the same member again
 - ▷ Without replacement: set the name aside. Each draw shrinks the sample
- ▶ Each vote can be yea or nay no matter what happened before, so we sample with replacement

Tallying the Results

```
table(repeat_trials)
```

```
repeat_trials  
NAY YEA  
  3   7
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- ▶ But we told `sample()` that yea and nay are equally likely (`prob = c(0.5, 0.5)`)

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- ▶ But we told `sample()` that yea and nay are equally likely (`prob = c(0.5, 0.5)`)
- ▶ So why 70–30 and not 50–50? To answer that, we must discuss independence

Independence



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- ▶ Trials are how we estimate that unobservable probability
- ▶ For trials to tell us anything, they have to meet some conditions. The first is independence: the outcome of one trial can't affect the outcome of another

Draws That Aren't Independent

- ▶ Back to the hat. In the 118th Congress, the Rules Committee had 9 majority (Republican) and 4 minority (Democratic) members

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- ▶ The first draw changed the probabilities for the second. These draws are not independent
- ▶ If draws are truly independent, the results of a trial tell us about the underlying probability of each outcome

So Why 70–30?

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- ▶ Our 10 votes were independent: `replace = T`, and no vote changes the next one's probabilities
- ▶ We still got 7 yeas and 3 nays
- ▶ In short: the trial was too small

The Law of Large Numbers

- ▶ Even when the true probability is $\{0.5, 0.5\}$, a small trial can look nothing like it

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Trials That Are Too Small

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- ▶ A one-vote trial: see a yea, and you'd conclude the probability is $\{1, 0\}$. Any nay after that would surprise you
- ▶ A two-vote trial:

```
sample(possible_outcomes, 2, replace = T, prob = c(0.5, 0.5))
```

```
[1] "YEA" "YEA"
```

- ▶ Two yeas! This trial says the member always votes yea

- ▶ What if we run the two-vote trial 10 times?

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```
trial_two_votes <- tibble(trial = 1:10) |>
  rowwise() |>
  mutate(outcome = list(sample(possible_outcomes, 2, replace = T,
                              prob = c(0.5, 0.5)))) |>
  unnest_wider(outcome, names_sep = "_") |>
  rename("vote_1" = outcome_1, "vote_2" = outcome_2)

trial_two_votes
```


Ten Two-Vote Trials

- ▶ What if we run the two-vote trial 10 times?

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trial_two_votes
```

- ▶ `rowwise()` draws a fresh pair of votes for each trial; `unnest_wider()` spreads each pair into two columns

Ten Two-Vote Trials

```
# A tibble: 10 x 3
  trial vote_1 vote_2
  <int> <chr>   <chr>
1     1    NAY    YEA
2     2    NAY    YEA
3     3    NAY    NAY
4     4    NAY    NAY
5     5    NAY    NAY
6     6    NAY    NAY
7     7    NAY    YEA
8     8    YEA    YEA
9     9    YEA    NAY
10    10    NAY    NAY
```

- ▶ 6 of our 10 trials (60%) came out unanimous: two yeas or two nays

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3     3  NAY    NAY
4     4  NAY    NAY
5     5  NAY    NAY
6     6  NAY    NAY
7     7  NAY    YEA
8     8  YEA    YEA
9     9  YEA    NAY
10    10  NAY    NAY
```

- ▶ 6 of our 10 trials (60%) came out unanimous: two yeas or two nays
- ▶ Each of those trials would badly mislead us about this member

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# A tibble: 10 x 3
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3     3  NAY    NAY
4     4  NAY    NAY
5     5  NAY    NAY
6     6  NAY    NAY
7     7  NAY    YEA
8     8  YEA    YEA
9     9  YEA    NAY
10    10  NAY    NAY
```

- ▶ 6 of our 10 trials (60%) came out unanimous: two yeas or two nays
- ▶ Each of those trials would badly mislead us about this member
- ▶ How can we be confident a trial reflects the true probabilities?

- ▶ The Law of Large Numbers: when a population of independent observations has a finite mean, then as the number of observations drawn increases, the mean of the observed values approaches the mean of the population

The Law of Large Numbers

- ▶ The Law of Large Numbers: when a population of independent observations has a finite mean, then as the number of observations drawn increases, the mean of the observed values approaches the mean of the population
- ▶ In other words: the more votes we observe, the closer we get to the true probabilities

A Function for Repeated Trials

- ▶ We'll run the same trial at five sizes, so we wrap it in a function of `n`, the number of votes:

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```
plot_votes <- function(n) {  
  tibble(outcome = sample(possible_outcomes, n, replace = T,  
                          prob = c(0.5, 0.5))) |>  
  tabyl(outcome) |>  
  ggplot(aes(y = outcome, x = percent)) +  
  geom_col() +  
  geom_point(aes(x = c(0.5, 0.5)), size = 3, colour = "#E21833") +  
  theme_minimal() +  
  scale_x_continuous(limits = c(0, 1))  
}
```


A Function for Repeated Trials

- ▶ We'll run the same trial at five sizes, so we wrap it in a function of `n`, the number of votes:

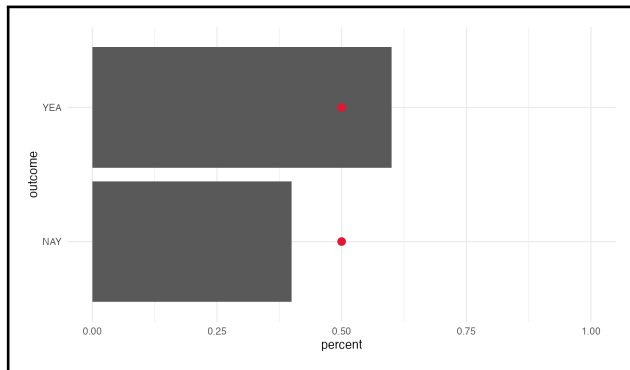
```
plot_votes <- function(n) {  
  tibble(outcome = sample(possible_outcomes, n, replace = T,  
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}
```

- ▶ `tabyl()` (from `janitor`) counts each outcome and its share. The red dots sit at 0.5: the true probability we're aiming for

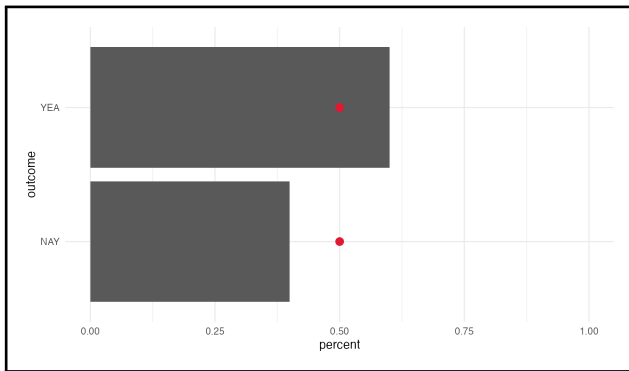
```
plot_votes(10)
```

10 Votes

```
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```



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```

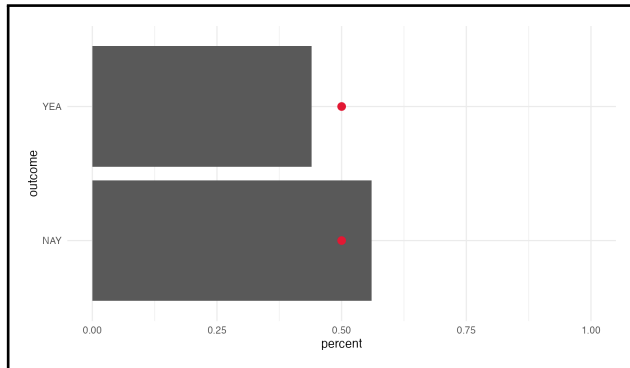


- ▶ 6 yeas, 4 nays (60–40). A different set of 10 votes from before gives a different answer

```
plot_votes(100)
```

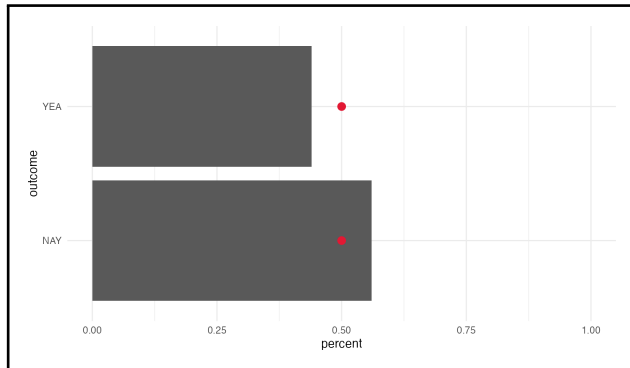
100 Votes

```
plot_votes(100)
```



100 Votes

```
plot_votes(100)
```



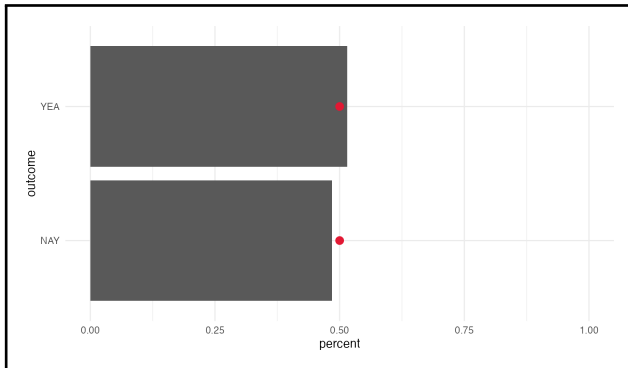
► 44 yeas, 56 nays. Closer, but still a noticeable gap

1,000 Votes

```
plot_votes(1000)
```

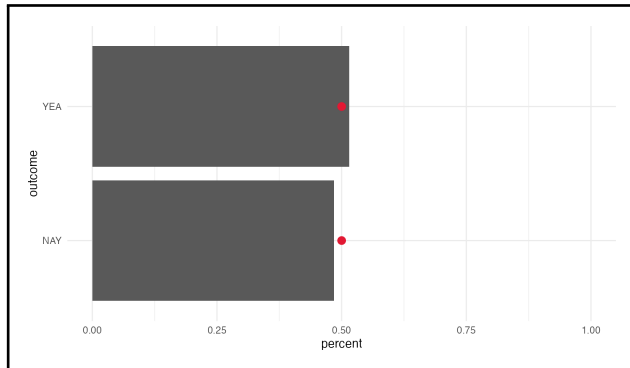

1,000 Votes

```
plot_votes(1000)
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1,000 Votes

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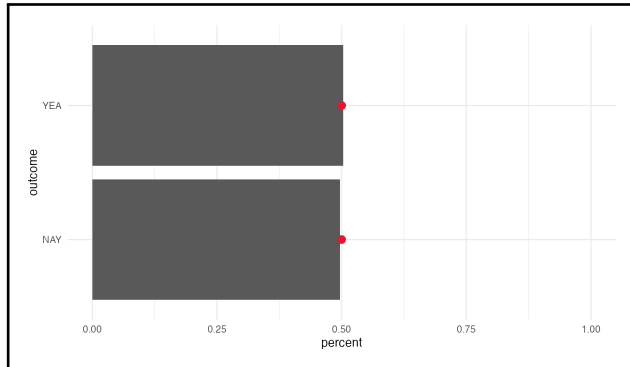
► 515 yeas, 485 nays (51.5%–48.5%)

10,000 Votes

```
plot_votes(10000)
```

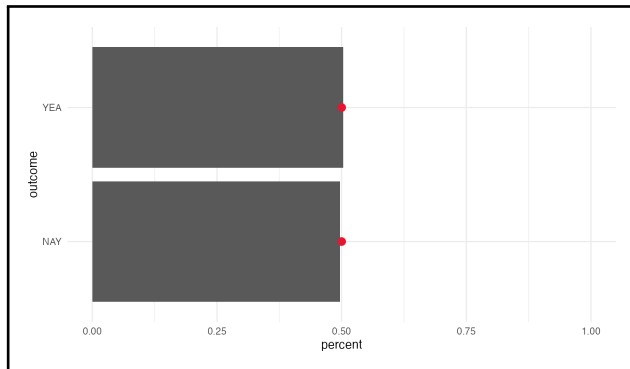
10,000 Votes

```
plot_votes(10000)
```



10,000 Votes

```
plot_votes(10000)
```



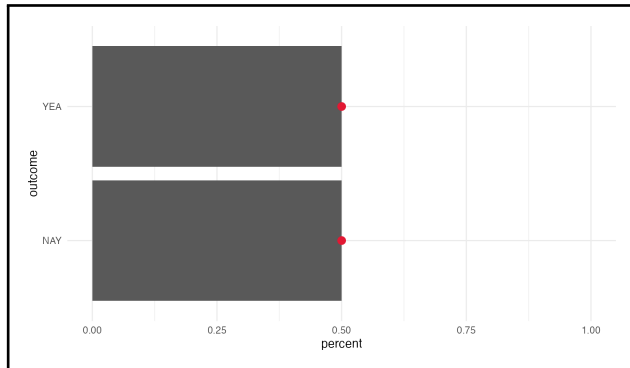
► 5,035 yeas, 4,965 nays (50.35%–49.65%)

100,000 Votes

```
plot_votes(100000)
```

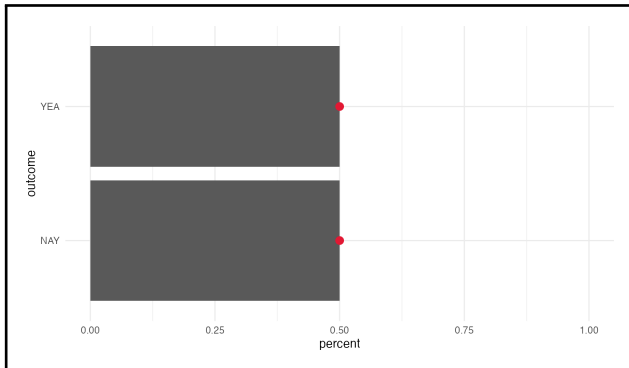
100,000 Votes

```
plot_votes(100000)
```



100,000 Votes

```
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```



- ▶ 49,975 yeas, 50,025 nays (49.975%–50.025%). The bars sit on the red dots.

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- ▶ Not always smoothly: 10 votes and 100 votes were both off by several points. Large numbers are what make the pattern reliable

So, YEA or NAY?

- ▶ Each time we increase the number of votes, we get closer to the true probabilities we gave `sample()`
- ▶ Not always smoothly: 10 votes and 100 votes were both off by several points. Large numbers are what make the pattern reliable
- ▶ Back to the whip count: for a truly 50/50 (“median”) member, neither guess gives you an edge. Both outcomes are equally likely

Conditional Probability

Case Study: Who Gets Laws Passed?

- ▶ Pick a member of the 118th House (2023–2024) at random. What is the chance that at least one bill they sponsored became law?

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Case Study: Who Gets Laws Passed?

- ▶ Pick a member of the 118th House (2023–2024) at random. What is the chance that at least one bill they sponsored became law?
- ▶ We'll use the Center for Effective Lawmaking (CEL) data from the past few labs
- ▶ We'll count a bill as becoming law if it passed on its own or its language was folded into another bill that became law

Preparing the Data

```
cel <- read_excel("CELHouse93to118.xlsx", guess_max = 20000)

house118 <- cel |>
  rename(
    congress = "Congress number",
    name = "Legislator name, as given in THOMAS",
    majority = "1 = majority party member",
    laws = "Total bills becoming law (inc in other leg vehicles)"
  ) |>
  filter(congress == 118) |>
  select(name, majority, laws) |>
  mutate(any_law = if_else(laws > 0, 1, 0))
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  select(name, majority, laws) |>
  mutate(any_law = if_else(laws > 0, 1, 0))
```

- ▶ `any_law` is 1 if at least one of the member's bills became law, 0 otherwise.
(`guess_max = 20000`: same reason as last lab)

```
house118
```

```
house118
```

```
# A tibble: 448 x 4
  name          majority laws any_law
  <chr>          <dbl> <dbl>   <dbl>
1 Adams, Alma      0      0       0
2 Aderholt, Robert 1      2       1
3 Aguilar, Pete    0      2       1
4 Alford, Mark     1      0       0
5 Allen, Rick      1      2       1
# i 443 more rows
```

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house118
```

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4 Alford, Mark         1     0       0
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# i 443 more rows
```

- (Output trimmed to 5 rows.) 448 members: everyone who served in the 118th House, including those who left or joined mid-session.

- ▶ On what share of members did at least one bill become law?

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```
tabyl(house118, any_law)
```

- ▶ On what share of members did at least one bill become law?

```
tabyl(house118, any_law)
```

any_law	n	percent
0	249	0.5558036
1	199	0.4441964

The Overall Probability

- ▶ On what share of members did at least one bill become law?

```
tabyl(house118, any_law)
```

any_law	n	percent
0	249	0.5558036
1	199	0.4441964

- ▶ $P(\text{law}) \approx 44\%$. If that's all we know, our best guess for a random member is that no law passed

- ▶ But the majority party controls committees and the floor. What if we know the member is in the majority?

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```
house118 |>  
  filter(majority == 1) |>  
  tabyl(any_law)
```

- ▶ But the majority party controls committees and the floor. What if we know the member is in the majority?

```
house118 |>  
  filter(majority == 1) |>  
  tabyl(any_law)
```

any_law	n	percent
0	103	0.4517544
1	125	0.5482456

- ▶ $P(\text{law} \mid \text{majority}) \approx 55\%$. Now our best guess flips to a law passed

```
house118 |>  
  filter(majority == 0) |>  
  tabyl(any_law)
```

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any_law	n	percent
0	146	0.6636364
1	74	0.3363636

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0	146	0.6636364
1	74	0.3363636

► $P(\text{law} \mid \text{minority}) \approx 34\%$

```
house118 |>  
  filter(majority == 0) |>  
  tabyl(any_law)
```

any_law	n	percent
0	146	0.6636364
1	74	0.3363636

- ▶ $P(\text{law} \mid \text{minority}) \approx 34\%$
- ▶ Knowing just one feature of the member moves our prediction from 34% to 55%

- ▶ “|” reads “given.” Filtering to the majority restricts the denominator to majority members only:

$$P(\text{law} \mid \text{majority}) = \frac{P(\text{law and majority})}{P(\text{majority})} = \frac{125/448}{228/448} = \frac{125}{228} \approx 0.548$$

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- ▶ Conditional probability uses everything we know about the circumstances of an event to update how likely we think it is
- ▶ Careful: this is prediction, not causal evidence. Majority members also chair every committee and differ in other ways. Is it the majority, or something that comes with it?

Two Ways to Think About Probability

- ▶ Frequentist: probability is the share of times an event happens over infinitely many identical trials. “A 50% chance of yea” means 50% yeas if the vote were repeated forever. We’ve been frequentists all session

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- ▶ Bayesian: probability is your belief about how likely an event is. It starts from what you already know (a prior) and is updated with each new observation

Two Whips, One New Member

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- ▶ The Frequentist won't predict anything yet: they want to see the member cast several votes first
- ▶ The Bayesian starts with a prior: with no other information, a 50/50 chance seems like a reasonable starting point
- ▶ Behind the scenes, we simulate a member who votes with the party 70% of the time. Neither whip knows this

The First Two Votes

```
whip_outcomes <- c("WITH PARTY", "AGAINST PARTY")  
  
first_votes <- sample(whip_outcomes, 2, replace = T, prob = c(0.7, 0.3))  
first_votes
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[1] "WITH PARTY" "WITH PARTY"
```

- ▶ Frequentist: two votes aren't enough to say anything
- ▶ Bayesian: two loyal votes don't rule out a 50/50 member, but they do push the belief up: $(10 + 2)/(20 + 2) \approx 55\%$ with the party

The Next 100 Votes

```
next_100 <- sample(whip_outcomes, 100, replace = T, prob = c(0.7, 0.3))  
table(next_100)
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next_100  
AGAINST PARTY    WITH PARTY  
              29              71
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► Now 102 votes: 73 with, 29 against

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next_100 <- sample(whip_outcomes, 100, replace = T, prob = c(0.7, 0.3))  
table(next_100)
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next_100  
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- ▶ Now 102 votes: 73 with, 29 against
- ▶ Frequentist: $73/102 \approx 72\%$. Willing to predict “with the party,” but wants more votes before being confident


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next_100  
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```

- ▶ Now 102 votes: 73 with, 29 against
- ▶ Frequentist: $73/102 \approx 72\%$. Willing to predict “with the party,” but wants more votes before being confident
- ▶ Bayesian: $(10 + 73)/(20 + 102) \approx 68\%$. The prior still pulls the estimate toward 50%

The Next 1,000 Votes

```
next_1000 <- sample(whip_outcomes, 1000, replace = T,  
                    prob = c(0.7, 0.3))  
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AGAINST PARTY    WITH PARTY  
          287          713
```

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```
next_1000  
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           287           713
```

► Now 1,102 votes: 786 with, 316 against

The Next 1,000 Votes

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next_1000 <- sample(whip_outcomes, 1000, replace = T,  
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```

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```

- ▶ Now 1,102 votes: 786 with, 316 against
- ▶ Frequentist: $786/1102 \approx 71\%$

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- ▶ Now 1,102 votes: 786 with, 316 against
- ▶ Frequentist: $786/1102 \approx 71\%$
- ▶ Bayesian: $(10 + 786)/(20 + 1102) \approx 71\%$

Where the Two Whips End Up

Votes observed	With party	Frequentist	Bayesian
0	0	—	50%
2	2	—	55%
102	73	72%	68%
1,102	786	71%	71%

- ▶ With enough data, both converge on the truth (70%). The data overwhelm the prior

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Votes observed	With party	Frequentist	Bayesian
0	0	—	50%
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102	73	72%	68%
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- ▶ With enough data, both converge on the truth (70%). The data overwhelm the prior
- ▶ A real whip wouldn't start at 50/50: most members vote with their party most of the time. Good prior information is the Bayesian's advantage, and a bad prior is the risk

Why This Matters for Political Science

- ▶ We start with a parameter of interest: e.g., what share of Americans approve of the job Congress is doing?

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- ▶ This is statistical inference

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- ▶ The Law of Large Numbers: more observations get us closer to the truth
- ▶ Conditional probability updates predictions with what we know: 44% overall, 55% for majority members
- ▶ Frequentists and Bayesians interpret probability differently, but with enough data they converge

Center for Effective Lawmaking. 2025. "House Legislative Effectiveness Scores, 93rd–118th Congresses." <https://thelawmakers.org>.

Volden, Craig, and Alan E. Wiseman. 2014. *Legislative Effectiveness in the United States Congress: The Lawmakers*. New York: Cambridge University Press.

Questions?